

GNSS analysis report

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ABSTRACT

This paper presents the results of the GNSS signal analysis performed as part of the "Wireless and device-to-device communication security" project laboratory. It describes the data collection process, including where and how the signals were acquired, as well as the tools and methodologies used for processing and analysis. The study also investigates the impact of spoofing, emulated in software during post-processing, on the estimated GNSS solution.

1 METHODOLOGY

Data were collected using the GNSS Logger application (v3.1.1.2) on a Samsung Galaxy S22 Android smartphone. Post-processing and analysis were performed in MATLAB using scripts based on the modified Google GNSS measurement framework provided for the laboratory. The framework was used for pseudorange evaluation, receiver position estimation, and software-based spoofing emulation.

Two acquisition scenarios were considered: static and dynamic. The logging duration was set to 5 minutes for the static scenario and 10 minutes for the dynamic scenario. For the dynamic acquisition, a further 250 second acquisition was carried out to better isolate the specific case of interest. In both cases, data were collected in open-sky conditions, and power saving mode was disabled throughout the acquisitions.

Spoofing was emulated during post-processing through the modified MATLAB framework applied to the recorded GNSS measurements. The analysis focused on its impact on pseudorange observations and on the estimated receiver position.

2 RESULTS AND DISCUSSION

2.1 Open-sky analysis

We selected for signal acquisition the most open-sky area at the Politecnico di Torino surrounded by an urban environment as a baseline condition. While ideal open-sky Android measurements yield RMS errors under 10 m [4] and $C/N_0 > 40$ dB-Hz [5], the surrounding urban architecture introduces multipath effects. Consequently, we expect usable but degraded C/N_0 values (30–35 dB-Hz) and positioning errors on the order of 10–20 m [4] [5].

Based on the acquired data, the estimated positions form a compact cluster, with a 50% distribution radius of 17.8 m (Figure 1). The reference position used for the analysis is $(45.064860, 7.6576578, 241)$. We computed the horizontal positioning error e_i^{2D} as the square root of the sum of the squared North error $(\Delta N_i)^2$ and the squared East error $(\Delta E_i)^2$. The corresponding Root Mean Squared Error $RMSE_{2D}$ is then obtained as the square root of the mean of the squared horizontal errors $(e_i^{2D})^2$ over all N epochs.

These metrics describe the horizontal positioning accuracy of the GNSS solution. The average 2D error of 12.29 m represents the mean distance between the estimated positions and the reference

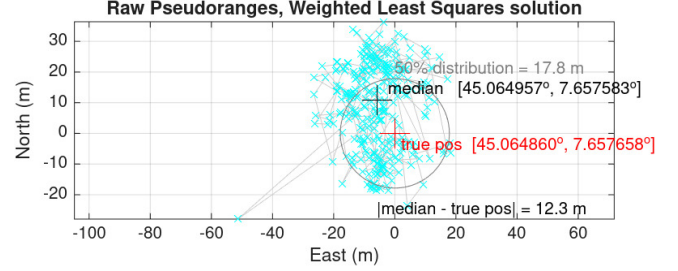


Figure 1: Estimated position cluster.

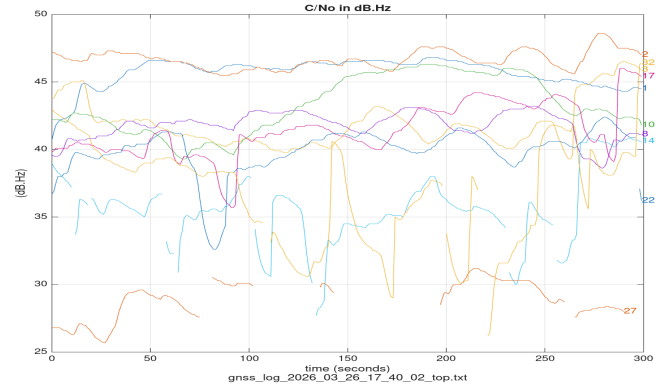


Figure 2: C/N_0 distribution.

position. The median 2D error of 17.8 m indicates the typical error level, meaning that 50% of the position estimates lie within 17.8 m of the true location. The standard deviation of 7.86 m shows that the error values are moderately spread, suggesting some variability in positioning performance over time. Finally, the RMSE of 19.49 m, which is higher than the mean error, highlights the presence of larger error peaks, likely caused by multipath effects or nearby obstacles. The following table reports, for each satellite, the mean value and the standard deviation of C/N_0 over the acquisition period. The overall mean C/N_0 is 40.61 dB-Hz, with a standard deviation of 5.02 dB-Hz. Contrary to our initial conservative expectations, this indicates a generally excellent signal quality with moderate variability across satellites and time.

Table 1: Mean and standard deviation of C/N_0 for each tracked satellite.

	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
σ	1.05	0.68	1.78	1.02	2.12	2.63	1.96	1.73	1.42	3.80
μ	45.68	46.67	41.93	41.80	43.29	35.57	41.46	39.87	28.89	36.72

2.2 Peculiar conditions data collection

In addition to the static baseline, a dynamic acquisition was conducted to characterize the route between the Torino Porta Susa railway station and the Politecnico di Torino campus. Analysis of the estimated path, shown in Figure 3, revealed significant localized drifts at specific locations compared to the ground truth. A closer examination of the environmental context at these points

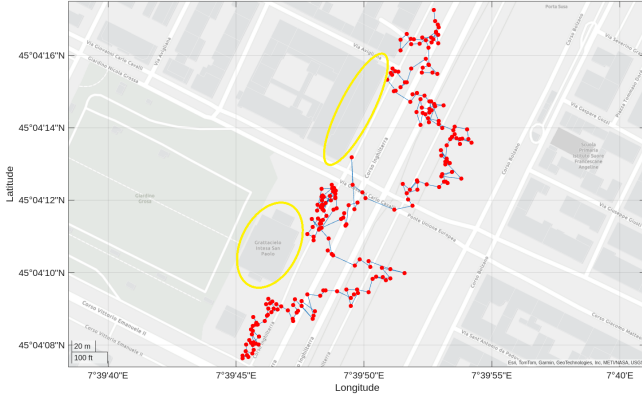


Figure 3: Estimated route from Porta SUSA train station to Politecnico.

indicates that these deviations coincide with the presence of high-rise structures, specifically Intesa Sanpaolo skyscraper and Città Metropolitana di Torino headquarter. Consequently, it is reasonable to hypothesize that multipath interference and Horizontal Dilution of Precision (HDOP) may constitute the primary sources of error within this urban environment.

As observed in the C/N_0 plot in Figure 4, it is noticeable that the signal power of some satellites exhibits sudden significant drops, going from the ideal open-sky levels of 40 dB-Hz to approximately 25 dB-Hz. This steep attenuation indicates a transition from open-sky conditions to Non-Line of Sight (NLOS) conditions, where the direct path is obstructed by skyscrapers or very high buildings, forcing the receiver to track reflected signals. Because of these reflections, the signal travels a longer distance than the direct path, introducing a positive bias in the pseudorange measurements.

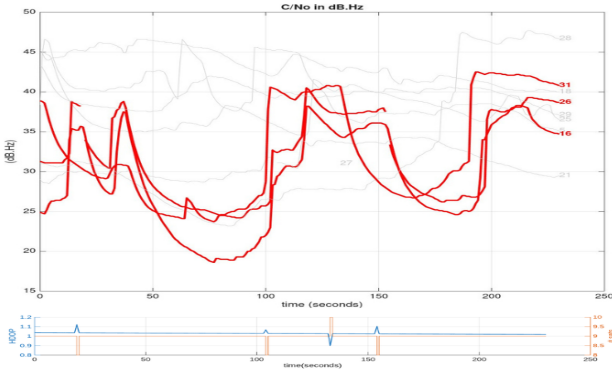


Figure 4: C/N_0 and HDOP from Porta SUSA train station to Politecnico.

Beyond signal power levels, HDOP is a useful metric for understanding the observed position drift. HDOP quantifies the impact of satellite geometry on horizontal accuracy; in urban environments like the one surrounding the Politecnico di Torino campus, tall buildings often block satellites in specific sectors of the sky, leading to poor geometric distribution. However, as demonstrated by the plot shown in Figure 4, HDOP remains remarkably stable throughout the acquisition, without significant peaks or fluctuations. The absence of geometric degradation implies that the recorded positional errors cannot be attributed to poor satellite constellation

geometry. Consequently, multipath interference and NLOS signals are the predominant sources of error in this environment.

2.3 Position Spoofing Analysis

We injected a false location by modifying the `spoof.position` parameter with the following coordinates, corresponding to the Intesa Sanpaolo skyscraper in Turin: Lat: 45.069626, Lon: 7.663147, Alt: 246 m; the attack was set to start at $t = 30$ s. These coordinates were selected based on the previous analysis.

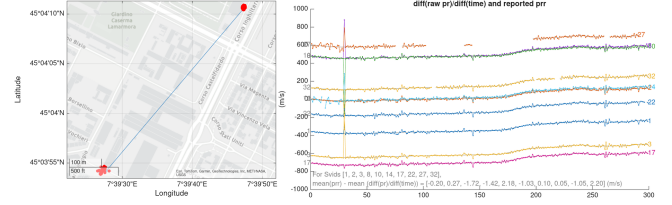


Figure 5: Map plot of spoof positions

The simulation produced the expected behavior for a basic step spoofing attack, as shown in Figure 5. The estimated position is shifted away from the true location at the start of the attack. Furthermore, by observing the position plot, it is noticeable that the points corresponding to the true trajectory are more scattered, while the points around the spoofed location are highly concentrated.

In the previous baseline analysis, the 50% distribution radius was 17.8 m, whereas in the spoofed case it decreases to about 2.5 m. However, this apparent improvement is not physically realistic. Based on our assumptions, such a small spread is unlikely for a real smartphone-based GNSS measurement, especially in the Intesa Sanpaolo area, where the surrounding urban environment is expected to introduce stronger multipath effects. In particular, reflections and diffraction caused by nearby buildings should increase the measurement variability rather than reduce it, as also suggested by the surrounding scenario shown in Figure 6.

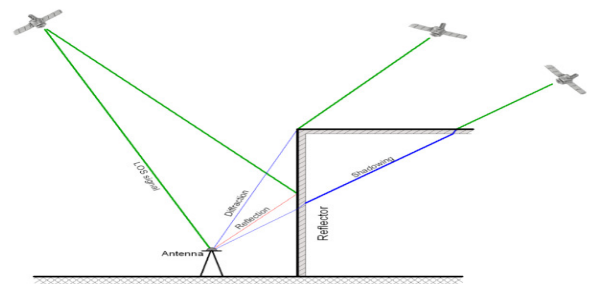


Figure 6: Type of received signals, image from [1]

Although the spatial shift on the map was expected, the spoofing effect is also evident in the internal receiver outputs. In particular, the PVT states plot in Figure 5 shows unrealistic velocity spikes. This occurs because the algorithm computes the solution epoch-by-epoch, effectively estimating velocity as the discrete time derivative of the position ($v \approx \frac{\Delta p}{\Delta t}$). Since the receiver is forced to jump to a distant position within Δt , the resulting estimated velocity becomes physically impossible and inconsistent with any realistic user motion.

2.4 Spoofing Detection Strategies

Based on the observed anomalies, several spoofing detection strategies can be implemented in a GNSS receiver. The **Kinematic Filtering**: As Modern receivers employ Kalman Filters, which predict the next state based on current dynamics. (based on measurement noise covariance matrix [3]). **Sensor Fusion**: Cross-referencing of GNSS data with internal IMUs (accelerometers/gyroscopes). The IMU would report zero physical acceleration, immediately flagging the GNSS “teleportation” as a spoofing event. **C/N₀ Monitoring**: As observed in our generated plots (Figure 5), there are no evident variations in the C/N₀ values during the attack. This occurs because our laboratory exercise performs a simulated spoofing at the data processing level (by manipulating the pseudoranges mathematically in MATLAB) rather than at the physical signal level. In contrast, during a real-world RF (Radio Frequency) spoofing attack, an attacker must broadcast fake signals at a higher power level to capture the receiver’s tracking loops. Therefore, while a sudden, simultaneous spike in C/N₀ across all channels would be a strong indicator of an actual RF attack, the absence of such a spike in our data clearly highlights the simulated, data-level nature of this specific scenario.

2.5 Spoofing Delay Analysis

To evaluate the effects of a temporal manipulation, we set the parameter `spoof.delay` to 3 ms while keeping the rest as the previous analysis.

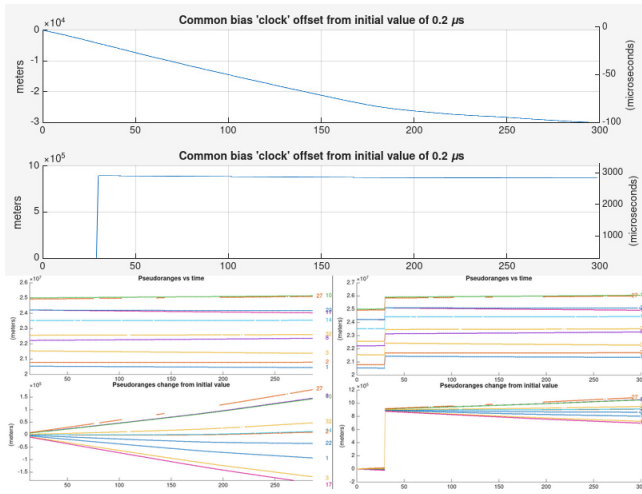


Figure 7: 3 ms delay on PVT states and comparison of raw pseudoranges.

Unlike the position spoofing analysis, where only the spatial coordinates were displaced, at $t = 30$ s, the PVT solution shifts instantaneously to the spoofed position and the receiver clock bias undergoes a jump of $3000 \mu\text{s}$, caused by a common-mode pseudorange shift of 9×10^5 m across all tracked satellites. The clock bias corruption follows from the 3 ms synthetic delay, which adds a fictitious distance $\Delta\rho = c \cdot \Delta t = 900,000$ m uniformly to all satellite signals. This dual corruption is further confirmed by the pseudorange evolution plots (Figure 7): prior to the attack, pseudoranges diverge gradually and independently for each satellite, reflecting normal orbital geometry changes. At $t = 30$ s, a simultaneous

common-mode jump is superimposed on all channels, after which the per-satellite drift resumes — a signature that uniquely identifies the presence of both a position displacement and a uniform delay injection. It is worth noting that the abrupt discontinuity observed at $t = 30$ s represents an idealised attack profile.

Attack realism. The simulated scenarios represent an idealised step-attack profile. To emulate a more plausible threat, an attacker would refine both the spatial and temporal characteristics of the spoofing. Spatially, the unrealistically compact distribution of the spoofed positions could be mitigated by adding Gaussian perturbations at each epoch, producing a PVT spread more consistent with real urban variability. Temporally, the attacker could gradually inject the position and delay parameters [2] rather than applying an abrupt jump. This smoothed transition would keep the apparent position variation and clock bias drift physically plausible, making the attack much less evident to derivative-based detection methods.

3 CONCLUSION

In real-world applications, incorrect position estimates in systems where positioning is critical can lead to financial losses, reputational damage for the operating company, and, in the most severe cases, risks to human health. The impact is even more critical for stationary timing receivers acting as a Primary Reference Time Clock (PRTC): an undetected 3 ms shift may silently corrupt the system clock and cause severe failures in timing-dependent infrastructures.

From a defensive perspective, understanding how a spoofed position may be detected also requires understanding which attack characteristics make spoofing less evident. Based on our analysis, concealing a spoofing attack is a complex task. It requires not only the injection of realistic forged positions, but also the ability to avoid detection mechanisms based on derivatives, temporal consistency, and other plausibility checks over time. In particular, reducing abrupt transitions is essential to limit detectable peaks in position and clock-related observables.

Our results suggest that position spoofing is significantly more complex than simple meaconing. While meaconing relies on the rebroadcast of real signals with delay, spoofing requires more advanced techniques to manipulate the navigation solution while remaining difficult to detect. Moreover, a realistic attack would also require dedicated hardware capable of transmitting counterfeit GNSS-like signals. Overall, the study highlights both the practical complexity of stealthy spoofing and the importance of robust detection strategies in safety- and timing-critical systems.

During the project, we analyzed several positioning scenarios, including indoor environments and multiple open-sky acquisitions. Among them, we selected the most representative cases, namely those in which interference effects were clearly observable. This allowed us to show how surrounding buildings can obstruct and reflect GNSS signals, thereby degrading the position estimate and providing several interesting insights, as illustrated in Figure 4.

APPENDIX

In this appendix, we report the custom MATLAB code snippets implemented to enhance the original ProcessGnssMeasScript.m. These additions were necessary to extract specific metrics (such as mean and standard deviation of the C/N₀), compute the tracking errors, and isolate the modified PVT states for the spoofing analysis.

```

%% C/N0 Metrics Extraction
meanCn0 = mean(gnssMeas.Cn0DbHz, 'all', 'omitnan')
;
meanCn0_t = mean(gnssMeas.Cn0DbHz, 2, 'omitnan');
meanCn0_sat = mean(gnssMeas.Cn0DbHz, 1, 'omitnan')
;
stdCn0 = std(gnssMeas.Cn0DbHz, 0, 'all', 'omitnan')
;
stdCn0_sat = std(gnssMeas.Cn0DbHz, 0, 1, 'omitnan')
;

disp('Standard Deviation C/N0 per satellite:');
disp(stdCn0_sat);
disp('Mean C/N0 per satellite:');
disp(meanCn0_sat);
disp('Overall Mean C/N0:');
disp(meanCn0);
disp('Overall Standard Deviation C/N0:');
disp(stdCn0);

%% 2. Single Satellite Plot Isolation
enableHighlight = 0; % 1 to enable, 0 to disable
targets = {'16', '26', '31'}; % Satellites to
highlight

if enableHighlight
    lines = findobj(gca, 'Type', 'line');
    texts = findobj(gca, 'Type', 'text');
    set(lines, 'Color', [0.8 0.8 0.8], 'LineWidth',
        0.5);
    set(texts, 'Color', [0.8 0.8 0.8]);

    for i = 1:length(targets)
        t = findobj(texts, 'String', targets{i});
        if isempty(t), continue; end

        for l = lines
            y = l.YData(~isnan(l.YData));
            if ~isempty(y) && abs(y(end) - t.
                Position(2)) < 1e-3
                set(l, 'Color', 'r', 'LineWidth',
                    2);
                set(t, 'Color', 'r', 'FontWeight',
                    'bold');
                break;
            end
        end
    end
end

%% Error Computation with Respect to True
Coordinates

% True coordinates (baseline reference)
lat_true = 45.064860; % [deg]

```

```

lon_true = 7.6576578; % [deg]
alt_true = 241.0; % [m]

% Extract valid estimates (non-NaN rows)
lat_est = gpsPvt.llaDegDegM(:,1);
lon_est = gpsPvt.llaDegDegM(:,2);
alt_est = gpsPvt.llaDegDegM(:,3);

valid = ~isnan(lat_est) & ~isnan(lon_est) & ~isnan
    (alt_est);
lat_est = lat_est(valid);
lon_est = lon_est(valid);
alt_est = alt_est(valid);

% Convert angular differences to meters
% 1 degree of latitude approx 111320 m
% 1 degree of longitude approx 111320 * cos(lat) m
R = 111320; % [m/deg]
dNorth = (lat_est - lat_true) * R;
% North error [m]
dEast = (lon_est - lon_true) * R * cosd(lat_true)
; % East error [m]
dUp = alt_est - alt_true;
% Vertical error [m]

% Horizontal error (2D) for each epoch
err_2D = sqrt(dNorth.^2 + dEast.^2);

% 3D error for each epoch
err_3D = sqrt(dNorth.^2 + dEast.^2 + dUp.^2);

```

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